

Name : Ricy Rifani Putra

Nim : A11.2025.16464

Matery : Queue

Output

```
▼ TERMINAL
● PS C:\MATKUL (TI) UDINUS SEM-2\ASD\Cpp> g++ stack-11/queue/queue.cpp -o queue
● PS C:\MATKUL (TI) UDINUS SEM-2\ASD\Cpp> ./stack-11/queue/queue.exe
Queue dengan class
[Sasuke | 0x255fceb9f10] --> [Sakura | 0]
[Naruto | 0x255fceba300] --> [Sasuke | 0x255fceb9f10] --> [Sakura | 0]
Calon Hokage urutan pertama adalah Naruto
Maaf, kapasitas penuh
[Kakashi | 0x255fceba1b0] --> [Naruto | 0x255fceba300] --> [Sasuke | 0]
Calon Hokage urutan pertama adalah Kakashi
❖ PS C:\MATKUL (TI) UDINUS SEM-2\ASD\Cpp> █
```

Source Code

```
stack-ll > queue > C++ queue.cpp
1 // Nama : Ricy Rifani Putra
2 // Nim : A11.2025.16464
3
4 #include <iostream>
5 using namespace std;
6
7 class Node {
8 public:
9     string nama;
10    Node *next;
11
12    Node() {
13        nama = "";
14        next = NULL;
15    }
16
17    Node(string nama) {
18        this->nama = nama;
19        this->next = NULL;
20    }
21 };
22
23 class Queue {
24 private:
25     Node *head;
26     Node *tail;
27     int kapasitas;
28
29 public:
30     Queue() {
31         head = NULL;
32         tail = NULL;
33         kapasitas = 100;
34     }
35
36     Queue(int kapasitas) {
37         head = NULL;
38         tail = NULL;
39         this->kapasitas = kapasitas;
40     }
41
42     bool isEmpty();
43     void enqueue(string nama);
44     void dequeue();
45     void enqueuePriority(string nama);
46     string peek();
47     int panjangQ();
48     bool isFull();
49     void cetak();
50     void dequeueBelakang();
51 };
52
53 bool Queue::isEmpty() {
54     return head == NULL;
55 }
```

```

57 int Queue::panjangQ() {
58     int count = 0;
59     Node *temp = head;
60     while (temp != NULL) {
61         count++;
62         temp = temp->next;
63     }
64     return count;
65 }
66
67 bool Queue::isFull() {
68     return panjangQ() >= kapasitas;
69 }
70
71 void Queue::enqueue(string nama) {
72     if (isFull()) {
73         cout << "Maaf, kapasitas penuh" << endl;
74         return;
75     }
76     Node *newNode = new Node(nama);
77     if (isEmpty()) {
78         head = newNode;
79         tail = newNode;
80     } else {
81         tail->next = newNode;
82         tail = newNode;
83     }
84 }
85
86 void Queue::enqueuePriority(string nama) {
87     if (isFull()) {
88         dequeueBelakang();
89     }
90     Node *newNode = new Node(nama);
91     if (isEmpty()) {
92         head = newNode;
93         tail = newNode;
94     } else {
95         newNode->next = head;
96         head = newNode;
97     }
98 }
99
100 void Queue::dequeue() {
101     if (isEmpty()) {
102         cout << "Queue kosong!" << endl;
103         return;
104     }
105     Node *temp = head;
106     head = head->next;
107     if (head == NULL) tail = NULL;
108     delete temp;
109 }

```

```

111 void Queue::dequeueBelakang() {
112     if (isEmpty()) {
113         cout << "Queue kosong!" << endl;
114         return;
115     }
116     if (head == tail) {
117         delete head;
118         head = NULL;
119         tail = NULL;
120         return;
121     }
122     Node *temp = head;
123     while (temp->next != tail) {
124         temp = temp->next;
125     }
126     delete tail;
127     tail = temp;
128     tail->next = NULL;
129 }
130
131 string Queue::peek() {
132     if (isEmpty()) return "";
133     return head->nama;
134 }
135
136 void Queue::cetak() {
137     if (isEmpty()) {
138         cout << "Queue kosong!" << endl;
139         return;
140     }
141     Node *temp = head;
142     while (temp != NULL) {
143         cout << "[" << temp->nama << " | " << temp->next << "]";
144         if (temp->next != NULL) cout << " --> ";
145         temp = temp->next;
146     }
147     cout << endl;
148 }

```

```

150 int main() {
151     cout << "Queue dengan class" << endl;
152     Queue hokage(3);
153
154     hokage.enqueue("Sasuke");
155     hokage.enqueue("Sakura");
156     hokage.cetak();
157
158     hokage.enqueuePriority("Naruto");
159     hokage.cetak();
160     cout << "Calon Hokage urutan pertama adalah " << hokage.peek() << endl;
161
162     hokage.enqueue("Boruto");
163     hokage.enqueuePriority("Kakashi");
164     hokage.cetak();
165     cout << "Calon Hokage urutan pertama adalah " << hokage.peek() << endl;
166
167     return 0;
168 }

```

**** Terimakasih ****